

The Future of Work – AI and Automation

MCIT – DEPI Round 4

Group: YAT684B_CAI4_DAT1_S5_DEPI4

Track: Data Analyst Specialist

Supervisor: Prof. Amal Mahmoud

Members	
1.	Helen Wageh
2.	Malak Emad
3.	Seif Eldeen
4.	Kareem Mohamed
5.	Tasneem Ayman

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Executive Summary

This project provides an end-to-end analysis of a global job market dataset to evaluate the impact of Artificial Intelligence across industries, job roles, and regions. The project uncovers key trends in AI adoption, salaries, employment patterns, and automation risk, presenting the findings through an interactive dashboard that supports informed decision-making.

Problem Statement

As Artificial Intelligence continues to reshape the workforce, there is a growing need to analyze its impact across industries, job roles, and regions. This project addresses that need by uncovering key trends and presenting them through an interactive analytical solution.

Project Objectives

- **Analyze AI's Impact:** Examine how Artificial Intelligence is reshaping employment trends, workforce dynamics, and the global job market.
- **Identify Market Trends:** Analyze AI adoption, salary patterns, and automation risk across industries, job roles, and geographic regions.
- **Support Talent Acquisition:** Provide organizations and recruiters with insights into in-demand roles, required skills, and emerging hiring trends.
- **Guide Career Development:** Help job seekers identify rewarding career paths, understand where opportunities are concentrated, and recognize the skills needed to remain competitive in an AI-driven job market.

- **Deliver Actionable Insights:** Present key findings through an interactive dashboard that supports effective data exploration and informed decision-making.

Project Workflow

1. Data Collection
2. Data Cleaning & Preprocessing
3. Data Modeling
4. Exploratory Data Analysis (EDA)
5. Data Transformation
6. KPI Development
7. Dashboard Design & Visualization
8. Insight Generation & Recommendations

Methodologies

Data Analytics Methodology - A structured end-to-end data analytics approach was followed, beginning with data collection and preparation, followed by data modeling, exploratory analysis, visualization, and insight generation.

Exploratory Data Analysis (EDA) - Exploratory techniques were applied to understand the dataset, identify trends, detect anomalies, and uncover relationships between variables before developing the final dashboard.

Tools and Technologies

Tool Used	Purpose
Excel - PowerPivot	Data cleaning, preprocessing, validation, and initial analysis. modeling large datasets beyond Excel's standard limitations.
SQL	Database design, data modeling, querying, and data transformation.
Python	Exploratory Data Analysis (EDA), statistical analysis, and data exploration.
Power BI	Interactive dashboard prototyping, data visualization, and business intelligence reporting.
Tableau	Interactive dashboard development, KPI visualization, and advanced data storytelling.

Use Case Scenarios

Recruiters & HR Professionals - Leverage workforce insights to identify in-demand roles, required skills, and hiring trends across industries, supporting more effective recruitment and talent planning.

Job Seekers - Explore career opportunities by identifying high-demand, high-paying roles, understanding required AI skills, and tracking employment trends across regions and industries.

Business Decision Makers - Monitor AI adoption, salary trends, and automation risk to support strategic workforce planning and data-driven organizational decisions.

Researchers & Analysts - Analyze labor market patterns, evaluate the impact of AI on employment, and uncover trends that support research and policy development.

Challenges & Solutions

Large Dataset Handling - The project involved a large dataset that exceeded Excel's standard capabilities. This challenge was addressed by leveraging **Power Pivot** to efficiently manage, model, and analyze the data.

Tools Integration - Maintaining a seamless workflow across Excel, SQL, Python, Power BI, and Tableau required careful coordination to preserve data integrity throughout the analytics process.

Dashboards Design & Insight Organization - Presenting numerous KPIs and visualizations without overwhelming users was addressed by designing a clear, interactive, and user-friendly dashboard layout.

Project Deliverables and Outcomes

- Cleaned and preprocessed analytical dataset.
- Relational SQL database with optimized schema.
- Set of analytical SQL queries and Python-based EDA.
- Interactive Tableau, Power BI, and Excel dashboards with dynamic filters and KPIs.
- Comprehensive project documentation and analytical report.